2. (*a*) Explain with diagram the iterative waterfall model of software development. What are the deliverables that need to be produced at the end of each phase ? 4

(*b*) What are the relative advantages of using either the LOC or the function point metric to measure the size of a software product ? 4

Or

(*a*) What are the activities carried out during phases of Spiral model ? State its advantages over other models for SDLC. 4

(*b*) Discuss how the effort spent in different phase of iterative waterfall model are spread over time. 4

VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY (VSSUT), ODISHA

Even Mid Semester Examinationfor Academic Session 2023-24

COURSE NAME: B.Tech

SEMESTER:6th

BRANCH NAME: CSE

SUBJECT NAME: Software Engineering

FULL MARKS: 30

TIME: 90 Minutes

Answer All Questions.

The figures in the right hand margin indicate Marks. *Symbols carry usual meaning.*

Q1. Answer all Questions. [2 × 3]

a) Define software engineering. Differentiate between a program and a software. - CO1

b) Consider a software product with 5 tasks T1-T5. Duration of the 5 tasks (in days) are 15, 10, 12, 25 and 10 respectively. T2 and T4 can start when T1 is complete. T3 can start when T2 is complete. T5 can start when both T3 and T4 are complete. When is the latest start date of the task T3? What is the slack time of the task T4? - CO2

c) List characteristics of a bad SRS document with example. - CO2

Q2. [4+4]

a) Explain how the software design techniques have evolved in the past. - CO1

b) Explain the phases of waterfall model for software development with schematic diagram. List its major disadvantages.

OR

a) Explain the prototype software life cycle model with diagram. Discuss the applications of it. - CO1

b) Draw a labelled schematic diagram to represent the spiral model of software development. Is the number of loops of the spiral fixed? If yes, write down the number of loops of spiral, otherwise explain how the number of loops of spiral is determined.

VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY (VSSUT), ODISHA

Even Mid Semester Examination for Academic Session 2023-24

COURSE NAME: B.Tech | SEMESTER: 6th

BRANCH NAME: IT

SUBJECT NAME: Software Engineering

FULL MARKS: **30** | TIME: **90 Minutes**

Answer **All** Questions.

The figures in the right hand margin indicate Marks. *Symbols carry usual meaning.*

Q1.		Answer all Questions.	**[2 × 3]**
	a)	Define software engineering as per IEEE. Differentiate between a program and software.	- CO1
	b)	What constitute a bad SRS? Why SRS document is considered as a Black-box specification?	- CO2
	c)	What is Life Cycle Model? Why model the software life cycle?	- CO2
Q2.			**[4+4]**
	a)	Explain how the software design techniques have evolved in the past.	- CO1
	b)	Explain the phases of Iterative waterfall model for software development with schematic diagram. List its major disadvantages. What do you mean by phase containment error?	
		OR	
	a)	Distinguish between Process, project and products. List and explain the characteristics of a software process.	- CO1
	b)	Draw a labelled schematic diagram to represent the spiral model of software development. Is the number of loops of the spiral fixed? If yes, write down the number of loops of spiral; otherwise explain how the number of loops of spiral is determined. Why spiral model is considered to be a Meta model?	

Q4.			[8]
	-3	Which of the life cycle model you follow for developing software for each of the following application? Mention the reason behind the choice of a particular life cycle model: i) A well understood data processing application. (waterfall) ii) A new software product that would connect computer through satellite communications. Assume that your team has no previous experience in developing satellite communication software. (spiral) iii) A software product that would function as the controller of a telephone switching system. (prototype) iv) A new library automation software that would link various libraries in the city. (waterfall) v) A extremely large software that would provide, monitor and control cellular communication among its subscribers using a set of revolving satellites. (spiral) vi) A new text editor. (spiral) vii) A compiler for a new language. (waterfall) viii) An object oriented software development effort. (evolutionary)	- CO2

2. (*a*) Briefly explain the process of developing software using prototype model. Under what circumstances is it beneficial to develop a prototype model ? Give the advantages and disadvantages of using this model. 4

(*b*) Explain CFG. Draw a neat CFG for the following C function named find maximum. From the CFG determine the Cyclomatic complexity.


```
int find-maximum(int i, int j, int k){
      int max;
      if(i>j) then
if(i>k) then max=i;
else max=k;
else if(j>k) max=j;
else max=k;
return max;
}
```

4

Or

(*a*) Discuss the laws related to maintenance. The changes made to a software product to add some desired features can be considered as which type of maintenance ? Justify your answer. 4

(*b*) Explain how software cost estimation can be carried out using COCOMO model ? What is the limitation of cost estimation model ?

3. (*a*) Develop the SRS document for the following problem :

Consider the following Automobile Spare Parts Shop Automation Software (AAS) which is required by a retail automobile spare parts shop to automate various book-keeping activities associated with its business.

(*i*) The retail shop deals with a large variety of automobile spare parts procured from various manufacturers. The shop owner maintains different parts in wall mounted and numbered racks.

(*ii*) At the end of each day, the shop owner would request the computer to generate indents for the items which are out of stock. The computer should print out the parts description, the quantity required and the address of the vendor supplying the part. The

shop owner would have to simply put these printouts into envelopes and courier them to the address printed.

(*iii*) Whenever new supplies arrive, the shop owner should be able to update the inventory.

(*iv*) Whenever any sales occurs, the shop owner would enter the code number of the parts and the corresponding quantities sold. AAS should print out the cash receipt, maintain the cash book and adjust the inventory.

(*v*) The computer should also generate the revenue for any specified day and month, when queried by the owner. 3

1. Answer *all* questions : 2×5

(*a*) Do you agree with the statement : "Code" is the most important reuse artifact that can be used during software development ? Justify your answer.